

Class notes | Module 1

Lecture One | Life

What is life?

- Organisms that grow, reproduce, respond to stimuli, regulate their internal environment, and use external energy

What are atoms?

- Everything we can see is made of atoms
- **Structure of atoms**
 - **Outer shell:** electrons
 - **Inner shell:** protons and neutrons
- Atoms can have positive, negative, or neutral charge
- Electrons can only orbit their atoms in certain shells

What are molecules?

- When multiple atoms are bound together by attractive forces, they form a molecule
- **Covalent bond:** chemical bond where two atoms share a pair of electrons
 - Number of possible covalent bonds is determined by the number of empty spots in the atom's valence shell

Why is carbon important for life?

- A single carbon atom can form 4 covalent bonds, making it the backbone of many essential complex molecules
- Carbon helps build many essential amino acids and proteins, including DNA & RNA
- Silicon has a similar structure (ability to form 4 covalent bonds), so some scientists speculate that silicon-based life could be possible

What is a solvent?

- **Solvent:** liquid in which another substance dissolves
 - **Example:** salt dissolving in water

- Water is an excellent solvent because of its polarity (one side of the molecule is positive and the other side is negative)
- Ammonia has been proposed as an alternate solvent

What is abiogenesis?

- **Abiogenesis:** formation of life from non-living elements
 - We don't really understand how this happens
 - **Panspermia:** theory that life formed elsewhere in the universe and then travelled to earth
- **Steps**
 1. Production of organic molecules
 2. Replication of organic molecules (i.e., RNA replicates DNA)

Lecture Two | Drake Equation

$$N = R_* \cdot f_p \cdot n_e \cdot f_l \cdot f_i \cdot f_c \cdot L$$

N : number of civilizations in the Milky Way that can communicate across interstellar distances

R_* : rate at which stars form in the Milky Way

f_p : fraction of stars with planets

n_e : for stars with planets, the average number of potentially habitable planets

f_l : fraction of habitable planets with life

f_i : fraction of habitable planets with life that develop intelligent life

f_c : fraction of intelligent civilizations that develop the ability to communicate across interstellar distances

L : average lifetime of a communicating civilization

Lecture Three | Phylogeny

Aliens are commonly depicted as human-like

- This depiction (head, eyes, arms, legs, etc) is largely based on science fiction
- But appearance is based on evolutionary history, which aliens don't share

Tree of life (phylogeny)

- **Diagram showing the evolution of organisms from an original common ancestor**
 - Every organism on life is connected via the common ancestor
- **Evolution:** change over time in proportions of individual organisms that differ in one or more traits
- **Evidence for evolution**
 - Fossil records show convergence of species
 - DNA shows convergence of species
 - Geography shows overlap between species
 - Skeletons show common underlying structures (including vestigial structures and non-optimal structures)
 - Embryology shows historical structures (gills, tail, etc)

Lecture Four | Evolution

Natural selection

- **Definition:** organisms that possess more advantageous traits will be more likely to survive and reproduce
- **Requirements**
 1. Individuals in a population must exhibit variation in one or more traits
 2. Those traits must be heritable
 3. Possessing one or more of these traits must make individuals more likely to survive and reproduce
- **Appearance of design:** complex life looks carefully crafted
- **Origin of variation:** genetic mutation
 - Each human has roughly 100 mutations in their DNA
 - Mutations are completely random, but natural selection is not
 - **Dominant & recessive genes**
 - The traits encoded by **dominant genes** will be expressed as long as the offspring inherits it from **one** parent
 - The traits encoded by **recessive genes** will only be expressed if the offspring

- The traits encoded by **recessive genes** will only be expressed if the offspring inherits it from **both** parents

- **Misconceptions**

- Natural selection does not drive organisms to evolve what they “need”
 - Traits only develop due to mutations, which can be passed down based on desirability
 - Offspring cannot inherit characters that their parents acquire during their lives (fitness, injuries, etc)

Lecture Five | Extremophiles

Evolution & environment

- **Extremophiles:** organisms that thrive in extreme environments (often environments that are inhabitable for normal organisms)
- **E. coli Long Term Evolution Experiment**
 - Experiment that began in 1988
 - Scientists cultivated 12 different populations of E. coli
 - Each population was grown in a citrate-dense, low-glucose environment
 - E. coli thrives on glucose and cannot naturally process citrate
 - Samples were frozen every 75 days to preserve the genetic code
 - In 2003, one bacterium acquired a mutation that allowed it to process and digest citrate
- **Types of extremophiles**
 - **Thermophiles:** heat-biased organisms
 - **Psychrophiles:** cold-biased organisms
 - **Acidophiles:** acid-biased organisms (pH < 3)
 - **Alkaliphiles:** base-biased organisms (pH > 9)
 - **Piezophiles:** pressure-biased organisms
 - **Xerophiles:** dryness-biased organisms
 - **Halophiles:** salt-biased organisms
 - **Anaerobes:** low-oxygen-biased organisms
 - **Radioresistant:** radiation-biased organisms
- **Note on acids & bases**
 - **Acid:** substance that increases the concentration of H⁺ ions in a solution

- **Base:** substance that decreases the concentration of H^+ ions in a solution
- **LUCA (last unknown common ancestor)**
 - Likely had genes to survive hot, low-oxygen environments in hydrothermal vents at the bottom of the ocean

Lecture Six | Planetary surfaces

Four processes that shape surfaces

- **Impact cratering**
 - Bowl-shaped marks left by asteroids or comets after striking a planet's surface
 - Older planets should have more cratered surfaces
- **Erosion**
 - The wearing down or building up of geologic features by natural environmental processes (wind, ice, water)
 - Requires fluid that flows on the surface and a substantial atmosphere
- **Volcanism**
 - Eruption of lava from a planet's interior onto its surface
- **Tectonics**
 - Fracturing and movement of a planet's hard outer layer (lithosphere) into multiple plates
 - Tectonics may have brought about life on earth

Lecture Seven | Planetary orbits

Eccentricity: how much an ellipse/orbit deviates from a perfect circle

Astronomical Unit (AU): average distance between the earth and the sun

Kepler's Laws

- **Kepler's First Law:** the orbit of every object around the sun is an ellipse with the sun at one focus/point
- **Kepler's Second Law:** planets move faster when closer to the sun, and slower when further away
- **Kepler's Third Law:** the orbital period of an object is related to the semi-major axis of its orbit and the mass of whatever it is orbiting according to the following equation

- If...
 - The object is orbiting the sun
 - The semi-major axis is measured in AU
 - The period P is measured in years
- Then the equation is just $P^2 = a^3$

The **orbital period** P of an object is related to the **semi-major axis** a of its orbit and the **mass** M of whatever it is orbiting by the following expression

$$P^2 = \frac{4\pi^2 a^3}{GM}$$

P is the **orbital period** (the time it takes for the object to complete 1 orbit)

a is the **semi-major axis** of the object's orbit

G is Newton's **gravitation constant** = $6.67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ (You do not need to memorize this number.)

M is the **mass** of whatever is being orbited

Lecture Eight | Newton's Las

Key terms

- **Velocity:** speed and direction of an object
 - An object's velocity always points in the direction it's moving
- **Acceleration:** change in velocity (via speed, direction, or both)
 - If an object's acceleration points in the same direction as its velocity, the object speeds up
 - If an object's acceleration points in the opposite direction as its velocity, the object slows down
 - If an object's acceleration points perpendicular to its velocity, the object's speed stays the same, but it changes direction
- **Force:** push or pull that one object exerts on another
 - **Unit:** Newton (N)

- **Finding net force**

- Force, like velocity and acceleration, has direction
- When two forces acting on an object point in the same direction, you add them
- When two forces acting on an object point in opposite directions, you subtract them

Newton's Laws

- **Newton's First Law:** an object moves at a constant velocity unless acted on by a force
 - **Newton's Second Law:** an object of mass M , experiencing a force F , will experience acceleration given by $F = ma$
 - **Newton's Third Law:** if Object 1 exerts a force on Object 2, Object 2 exerts an opposite but equal force on Object 1
 - **Newton's Law of Gravitation:** the gravitational force between two objects depends on the product of the masses of the objects and the distance between them
- Gravitational force is always attractive



Newton's Law of Gravitation

The **gravitation force** between two objects depends on the product of the **masses of both objects (M and m)** and the **distance r between the objects**

$$F_{grav} = \frac{GMm}{r^2}$$

F_{grav} is the strength of the **gravitational force** between the two objects

G is Newton's **gravitation constant** $= 6.67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ (You do not need to memorize this number.)

M and m are the **masses** of the two objects

r is the **distance** between the objects